

A unique hypergraph decomposition at every prime order

Ziang Ni (ziangni-sys)

10 October 2026

Disproof of conjecture 00000008437. The conjecture claims double-exponential numbers of decompositions of a complete t -uniform hypergraph into specified isomorphism classes when the number v of vertices is prime. It imposes no exclusion of single-edge block classes. That permitted class has exactly one decomposition at every order.

Take $t = 2$. The complete 2-uniform hypergraph $K_v^{(2)}$ has vertex set $V = \{0, \dots, v - 1\}$ and edge set

$$E = \{e \subseteq V : |e| = 2\}.$$

Specify the block isomorphism class H consisting of a single 2-edge on its two incident vertices. Each copy is indexed by one edge $e \in E$, and its edge set is $\{e\}$. All such blocks are isomorphic to H ; unused ambient vertices are not part of a block's vertex support.

A standard unordered decomposition is a collection $\mathcal{D}$ of edge-disjoint copies of H covering all of E . Coverage forces the block $\{e\}$ to occur for each $e \in E$: any block containing e has only one edge and hence must equal $\{e\}$. Conversely every block of $\mathcal{D}$ has that form. Therefore necessarily

$$\mathcal{D} = \{\{e\} : e \in E\}.$$

This collection is indeed a decomposition: its singleton edge sets are pairwise disjoint and their union is E . Thus the exact number of decompositions is

$$N_H(v) = 1 \quad \text{for every } v.$$

In particular this holds for every prime v , including arbitrarily large primes. A constant function cannot have double-exponential growth along the primes. For example, for every positive c , the lower bound $N_H(v) \geq \exp(\exp(cv))$ fails at every positive prime; more generally N_H does not even tend to infinity. This refutes the first conjunct, so no assertion about automorphism fixed-point spectra is needed.

Scope. This uses an admissible specified isomorphism class in the actual complete hypergraph. A theorem restricted to nontrivial multi-edge blocks would require a hypothesis absent from both source languages. If decompositions were artificially ordered, the count would instead be $\binom{v}{2}!$; even that is bounded by $\exp(O(v^2 \log v))$ and is not double-exponential in v . The usual unordered convention gives the stronger exact count above.

Lean coverage. The project represents actual edges as two-element subsets of $\text{Fin}(v)$, blocks as finite edge sets, and decompositions as unordered finite families satisfying the one-edge block condition, coverage, and pairwise disjointness. It proves existence and uniqueness, the exact cardinality one, and counterexamples at arbitrarily large primes. Lean 4.19.0 and Mathlib are publicly pinned; theorem audits use only standard logical axioms.